

The empirical recurrence for OEIS A198638, proved by a two-line transfer matrix

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Abstract

OEIS A198638 counts arrays in which every cell is constrained by its neighbours on *both* sides, and which are moreover written in canonical form (“values introduced in row major order”). Neither feature is visible to a one-line transfer matrix: the cell condition reaches the row above and the row below, and the canonical-form clause is not local at all. Both are dealt with. What is counted is equality patterns, so $3!a(n) = \sum_i \binom{3}{i} D_{3-i} L_i(n)$ with D_m the derangement numbers and L_i the count over an i -letter alphabet; and each L_i is a walk count whose vertices are ordered pairs of consecutive rows. Relabelling-invariance makes the chain strongly lumpable, cutting 98 pairs down to $S = 23$ states. The entry’s empirical recurrence of order 2, contributed by R. H. Hardin, then follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A198638 is “Number of $2n \times 2$ arrays with values introduced in row major order and each element equal to an even number of horizontal and vertical neighbors.”. It has offset 1 and begins

$$4, 36, 376, 3936, 41216, 431616, 4519936, 47333376, \dots$$

The entry carries the following comment:

Empirical: $a(n) = 12a(n-1) - 16a(n-2)$ for $n > 3$.

As of the “Last modified” line on the live entry (May 15 2018 06:42:06, revision 9) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 12a(n-1) - 16a(n-2) \tag{1}$$

for all $n > 3$.

2 The condition, and what the canonical-form clause counts

The arrays have 2 columns and $L = 2n + 0$ rows. Write $x_{t,u}$ for the entry in row t and column u , with the neighbour offsets

$$\mathcal{N} = \{(-1, 0), (1, 0), (0, -1), (0, 1)\}$$

written as (row shift, column shift), the requirement is

$$\#\{(d_1, d_2) \in \mathcal{N} : (t + d_1, u + d_2) \text{ lies in the array and its entry is equal to } x_{t,u}\} \text{ is even} \tag{2}$$

for every cell (t, u) .

The condition is stated purely in equalities between entries, so it is unchanged by any relabelling of the values: it is a property of the *equality pattern* — the partition of the cells into classes of equal entries — and not of the values themselves. The clause “values introduced in row major order” selects exactly one array from each relabelling class, so with $K = 3$ available values the entry counts admissible patterns with at most K classes.

Lemma 1. *Let $L_i(n)$ count the arrays over an alphabet of i letters that satisfy the cell condition, with no canonical-form clause, and $N_j(n)$ the admissible patterns with exactly j classes. Then $L_i = \sum_j N_j i(i-1)\cdots(i-j+1)$ and*

$$3! \sum_{j \leq 3} N_j(n) = \sum_{i=0}^3 \binom{3}{i} D_{3-i} L_i(n), \quad D_m = m! \sum_{t=0}^m \frac{(-1)^t}{t!}.$$

Proof. A labelled array satisfying the condition is a pattern together with an injection of its classes into the alphabet, and a pattern with j classes admits $i(i-1)\cdots(i-j+1)$ of them; that is the first identity. Writing $i(i-1)\cdots(i-j+1) = j! \binom{i}{j}$ it reads $L_i = \sum_j (j! N_j) \binom{i}{j}$, so binomial inversion gives $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$. Summing over $j \leq 3$ and exchanging the order of summation, the coefficient of L_i is $\sum_{t \leq 3-i} (-1)^t / (i! t!)$, which multiplied by $3!$ is $\binom{3}{i} (3-i)! \sum_{t \leq 3-i} (-1)^t / t! = \binom{3}{i} D_{3-i}$. $\square$

3 Each L_i is a walk count on pairs of rows

Fix i and let $V_i = \{0, \dots, i-1\}^2$. Because (2) at a cell of row t involves rows $t-1$, t and $t+1$, a single row is not enough state. Take the ordered pairs $(p, c) \in V_i \times V_i$ as vertices and declare $(p, c) \rightarrow (c, x)$ an edge when every cell of the middle row c satisfies (2) with p above it and x below it. Let P_i be the adjacency matrix, $\mathbf{s}_i$ the indicator of the pairs (p, c) whose first row p satisfies the condition with no row above, and $\mathbf{e}_i$ the indicator of the pairs whose second row c satisfies it with no row below.

Lemma 2. *For $L \geq 2$, the arrays of L rows satisfying the condition are exactly the walks $(r^{(1)}, r^{(2)}) \rightarrow (r^{(2)}, r^{(3)}) \rightarrow \dots \rightarrow (r^{(L-1)}, r^{(L)})$ of length $L-2$ that start at a vertex marked by $\mathbf{s}_i$ and end at one marked by $\mathbf{e}_i$, so $L_i = \mathbf{s}_i^\top P_i^{L-2} \mathbf{e}_i$.*

Proof. A walk of that form is exactly a sequence of rows $r^{(1)}, \dots, r^{(L)}$, since consecutive vertices overlap in one row. In such a walk the condition at the cells of row t is tested once, at the step leaving $(r^{(t-1)}, r^{(t)})$, and with the correct neighbours $r^{(t-1)}$ and $r^{(t+1)}$; the cells of row 1 are tested by $\mathbf{s}_i$, which uses the condition with the row above absent, and those of row L by $\mathbf{e}_i$. So every cell is tested exactly once and with its true neighbourhood, which is the claim. $\square$

Lemma 3. *Write $\pi(p, c)$ for the equality pattern of the pair. If $\pi(p, c) = \pi(p', c')$ then for every pattern Q the two vertices have the same number of successors of pattern Q , and $\mathbf{s}_i, \mathbf{e}_i$ take the same value at both. Hence the chain is strongly lumpable over π and L_i is computed on the quotient, whose states are the set partitions of the $2 \cdot 2$ cells of a pair into at most i classes.*

Proof. Equal patterns means $(p', c') = (\sigma(p), \sigma(c))$ for a permutation σ of the alphabet. The edge condition and the two boundary conditions are statements about equalities between entries, so they are invariant under σ , and $x \mapsto \sigma(x)$ is a bijection between the successor sets preserving patterns. The number of pairs with a given pattern P is $e_P = i(i-1)\cdots(i-|P|+1)$, the number of injections of its classes into the alphabet, which supplies the weights on the quotient. $\square$

Assembling the 3 lumped chains block-diagonally into N of size $S = 23$, with the weights of Lemma 1 folded into the start vector $\mathbf{v}$ and the end-of-array indicator into $\mathbf{u}$,

$$3! a(n) = \mathbf{v}^\top N^{L-2} \mathbf{u}, \quad L = 2n + 0. \quad (3)$$

4 The criterion, and the computation

Let $q(t) = t^2 - \sum_i c_i t^{2-i}$ be the characteristic polynomial of (1) and put $G = N^2$, so that consecutive values of n advance the walk by 2 rows.

Lemma 4. *With $h_j = G^j N^o \mathbf{u}$ for the offset o that aligns the first admissible index,*

$$3! \left(a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top G^{j-2} \left(h_2 - \sum_i c_i h_{2-i} \right)$$

for the corresponding j . Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for every $m \geq 0$, where $w = h_2 - \sum_i c_i h_{2-i}$; and it is enough to check $m = 0, 1, \dots, S-1$.

Proof. Substitute (3) and factor. The scalar $3!$ is nonzero, so it does not affect vanishing. For the last clause, Cayley–Hamilton applied to G makes G^S an integer combination of $I, G, \dots, G^{S-1}$, so each $\mathbf{v}^\top G^m w$ with $m \geq S$ is the same combination of the earlier values; if those vanish, so do all the rest. $\square$

Theorem 5. *The recurrence (1) holds for every $n > 3$, which is the statement of Section 1.*

Proof. The lumped transition counts were built by taking one representative pair for each pattern, enumerating every row x over the alphabet, testing (2) on the middle row, and sorting the results by the pattern of the new pair. The vector w was formed by 2 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots$, again exactly. Every one of those integers vanishes from the index corresponding to $n = 3+1$ onwards, and the run was continued until $S+1$ consecutive zeros had been seen, which by Lemma 4 settles every larger m . $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry’s DATA: the right-hand side of (3), divided by $3!$, was evaluated at every one of the 18 published indices and agrees exactly, the quotient coming out an integer each time. This tests the modelling and not only the arithmetic — the inversion of Lemma 1, the reading of (2), the treatment of the first and last rows, and the alignment of the entry’s index with $L = 2n + 0$ would each show up as a mismatch.

Second, the lumped chain was checked against the unlumped one on the small shapes of this family: the walk counts over the full alphabet agree with $\sum_P e_P f_m(P)$ term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 4 was carried to the full Cayley–Hamilton bound rather than to a sampled prefix.

References

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